

Benjamin Roget

Software Engineer — Backend & Systems

Software engineer (co-op) focused on backend development and systems tooling. I build reliable, maintainable systems by combining rigorous automated testing with declarative infrastructure (NixOS, Docker). Seeking a 3-month software engineering internship in the US, starting June 2027.

📍 Villeneuve d'Ascq, France

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🔗 portfolio.qgroget.com

🐙 [Strange500](#) (GitHub)

🌐 [roget-benjamin](#) (LinkedIn)

Work **Software Engineer (Co-op)**

Numih France (DOGRH)

Sep 2025 – Present

- Architected a unified E2E test engine (Spring Boot, Selenium) running identically via Maven in CI and an async REST API.
- Implemented cold precondition checks failing invalid runs in ~50ms (vs 20s browser startup), cutting false-positive CI failures by ~70%.
- Designed automated data-isolation routines in Jenkins pipelines to eliminate concurrent test collisions on shared environments.
- Developed and deployed a core Spring Boot internal automation tool to production.

Fullstack Developer Intern

Ekeep-IT

Apr 2025 – Jun 2025

- Engineered full-stack features and RESTful endpoints using Java Spring Boot to interface backend services with database layers.
- Developed responsive user interfaces in Next.js (React) and delivered sprint user stories through Agile/Scrum team rituals.

Logistics Assistant

Flint Group

Jun 2023 – Aug 2024

Seasonal role — summers 2023 and 2024.

- Managed inventory with SAP, ran weekly stock counts, and tracked finished-goods flows.

Education

IMT Nord Europe, Institut Mines-Télécom

Master of Engineering in **Software Engineering**

Sep 2025 – Present

University Institute of Technology of Lille

Bachelor of Science in **Computer Science**

Sep 2023 – Jun 2025

Projects

min-btc-node

Built a Bitcoin node from scratch in Rust, implementing the P2P wire protocol (handshake, inventory, block headers) with zero protocol dependencies and concurrent peer parsing.

Rust Bitcoin Networking

nixos-config

Declared multi-host NixOS configurations and dotfiles with Nix flakes and Home Manager to make development environments versioned and fully reproducible across machines.

Nix NixOS Linux IaC

pixel-war

Architected a Web3 pixel-canvas dApp in Solidity with pull-over-push refund patterns to prevent reentrancy attacks, and on-chain ERC-721 snapshot minting.

Web3 Solidity JavaScript Blockchain

HomeLab Infrastructure

Provisioned a self-hosted server with Unraid and Docker running Jellyfin, Gitea, and Pi-hole to learn bare-metal provisioning and network routing hands-on.

Unraid Docker Linux Self-hosting

RayTracer-Rust

CPU ray tracer in Rust with Bounding Volume Hierarchy (BVH) spatial partitioning for $O(\log N)$ ray-primitive intersections, Phong shading, and reflections.

Rust Graphics Ray Tracing BVH

QGChat

Full-stack chat app in Java EE with Docker Compose deployment, SQL persistence, and a JavaScript frontend.

- Java
- Java EE
- Docker
- WebSocket
- SQL

ascii-cube-rs

Real-time ASCII 3D cube renderer compiled to WebAssembly via a Nix flake — powers the hero widget on this portfolio.

- Rust
- WebAssembly
- Nix

Skills

Languages & Systems

- Rust
- Java
- C
- Python
- TypeScript
- JavaScript
- WebAssembly

Backend & Web

- Spring Boot
- Java EE
- React
- Next.js
- Node.js
- Angular

DevOps & Infrastructure

- NixOS
- Linux
- Docker
- Podman
- Jenkins
- Traefik
- Git

Architecture & Quality

- Selenium
- JUnit 5
- Design Patterns
- Agile

Languages

French

Native

Spanish

Intermediate

English

Professional — TOEIC 980

Interests

Sport shooting

- national level

Swimming

Weightlifting

Running

Cycling